

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 15)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

01 . 1 Give **three** sources of mercury pollution.

[3 marks]

1

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2

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3

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01 . 2 Describe **three** methods used to control mercury pollution.

[3 marks]

1

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01 . 3 Explain why heavy metal wastes are stored under alkaline conditions.

[2 marks]

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01 . 4 State what is meant by synergism and give an example involving a heavy metal.

[2 marks]

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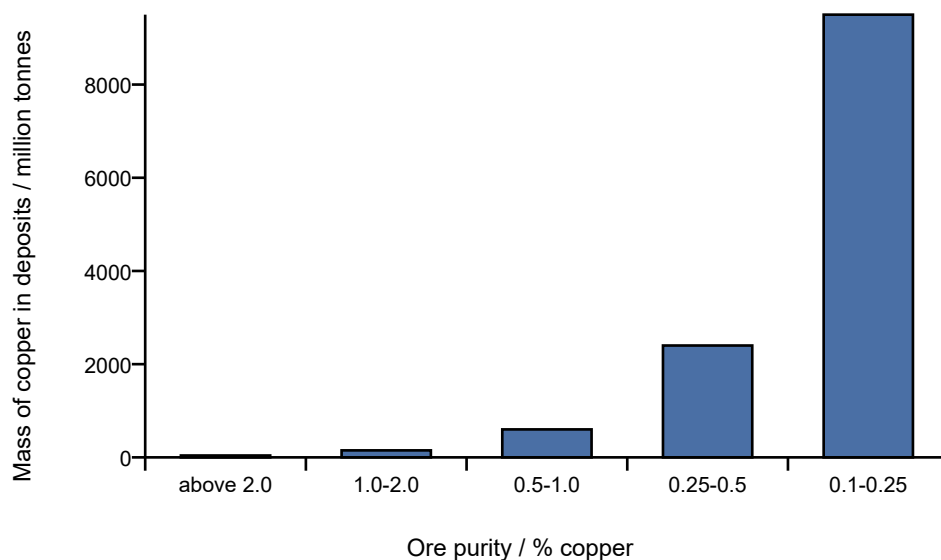
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10

0 2

Figure 1 shows the mass of copper contained in the Earth's crust at different ore purities, illustrating Lasky's principle.

Figure 1



0 2

1

State Lasky's principle.

[1 mark]

0 2

2

Use Figure 1 to calculate how many times more copper is contained in deposits of 0.25-0.5% purity than in deposits above 2.0%.

[1 mark]

Answer

times

0 2

3

Explain the difference between the **reserves** and the **resource** of a mineral.

[3 marks]

0 2 . 4

Explain, using Lasky's principle, why the ability to exploit slightly lower grade ores greatly increases reserves.

[2 marks]

0 2 . 5

Explain why the main problem for future mineral supplies is not the quantity that exists in the crust.

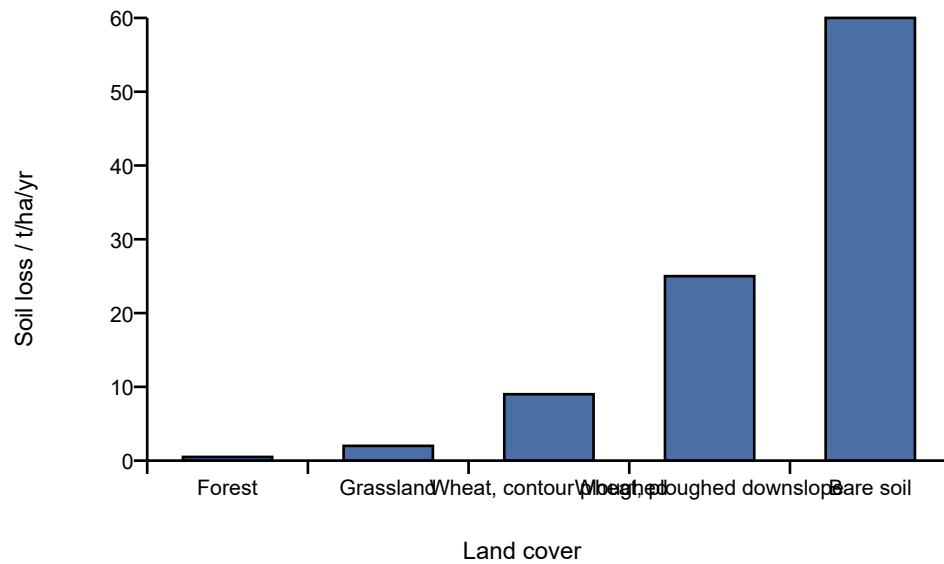
[3 marks]

10

03

Figure 1 shows the annual soil loss measured on experimental plots with different vegetation cover on a 10% slope in a region with heavy seasonal rain.

Figure 1



03

1 Describe the pattern shown in Figure 1.

[2 marks]

03

2 Explain **four** ways in which vegetation reduces the rate of soil erosion.

[4 marks]

03 . 3

Explain why ploughing up and down a slope causes more erosion than contour ploughing.

[2 marks]

03 . 4

Explain why the region's climate makes the soil vulnerable to **both** wind and water erosion.

[2 marks]

10

04 . 1

Which gas makes up about 21% of the dry atmosphere by volume? Tick **one** box.

[1 mark]

A Nitrogen

☐

B Oxygen

☐

C Argon

☐

D Carbon dioxide

☐

04 . 2

Explain why it is difficult to predict the effect of a human activity on atmospheric processes.

[2 marks]

04 . 3

Explain how winds transport both energy and water to areas such as the UK.

[2 marks]

5

0 5 Table 1 shows the mass of carbon in the main reservoirs of the carbon cycle.

Table 1

Reservoir	Mass of carbon / billion tonnes
Atmosphere (CO ₂)	700
Living organisms	470
Dead organic matter (soil)	3 700
Surface ocean waters	500
Deep ocean (dissolved)	35 000
Fossil fuels	10 000
Sedimentary carbonate rocks	20 000 000

0 5 . 1 Express the mass of carbon in the atmosphere in standard form, in tonnes.

[1 mark]

Answer tonnes

0 5 . 2 Calculate the percentage increase in atmospheric carbon that would result from burning 10% of the fossil fuel reservoir, assuming all the carbon remained in the atmosphere.

[2 marks]

Answer %

0 5 . 3 Explain why the sedimentary carbonate rocks, although by far the largest reservoir, have little effect on atmospheric CO₂ concentrations.

[2 marks]

0 5 . 4

Describe how carbon moves from the atmosphere into sedimentary carbonate rocks.

[3 marks]

0 5 . 5

Explain why aerobic respiration returns carbon to the atmosphere as CO_2 but anaerobic respiration returns it mainly as methane.

[2 marks]

10

0 6 . 1

Describe the main features of the Montreal Protocol (1987).

[3 marks]

06 . 2

For each of the following CFC uses, give the alternative process or material now used and one disadvantage of the alternative.

[4 marks]

Aerosol propellant

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Refrigerant

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06 . 3

Describe how waste CFCs from old refrigerators are disposed of safely.

[2 marks]

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06 . 4

State **one** reason why the ozone layer is not expected to recover fully for many decades.

[1 mark]

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10

0 7

Table 1 shows the bypass ratio and first year of use of jet engines used on four airliners.

Table 1

Engine	Aircraft	Bypass ratio	First use
RR Olympus	Concorde	0 : 1	Early 1950s
RR Conway	Boeing 707	0.25 : 1	Late 1950s
P&W JT9D	Boeing 747	5 : 1	1969
RR Trent 1000	Boeing 787	10 : 1	2011

0 7**1**

Explain how a high bypass ratio reduces engine noise.

[3 marks]

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0 7**2**

Describe **two** other features of aircraft design that reduce noise.

[2 marks]**1****2****0 7****3**

Explain how each of the following airport operating procedures reduces noise nuisance.

[4 marks]

Constant descent angle

.....

Night flight restrictions

.....

Noise-based landing charges / quota count

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Multiple runways

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0 7 . 4

Explain why aircraft are noisier on take-off than on landing.

[1 mark]

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10

0 8 . 1

Explain the locational requirements for harnessing wave power.

[2 marks]

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0 8 . 2

Describe how **one** named wave power device generates electricity.

[2 marks]

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0 8 . 3

Give **two** design problems that have delayed the development of wave power.

[2 marks]

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1

2

0 8 . 4 Compare the environmental impacts of tidal barrages, tidal lagoons and in-stream tidal turbines. **[4 marks]**

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10

0 9 A student investigated the effect of pH on the germination of cress seeds.

0 9 . 1 Explain why cress is a suitable species for this investigation but the results may not apply directly to natural ecosystems. **[2 marks]**

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0 9 . 2 Describe how the student should decide the range of pH values to test. **[2 marks]**

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09 . 3

Give **three** variables that should be standardised and explain how long the experiment should continue.

[4 marks]

09 . 4

The student wants to compare the effect of nutrient concentration on the growth of duckweed. Explain why leaf area is a better measure of growth than fresh mass.

[2 marks]

10

10 . 1

Evaluate the risks and benefits of the uses of ionising radiation, and explain the principles used to control exposure of workers and the public.

[9 marks]

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1 0 . 2

Give **one** reason why the risks and benefits of nuclear electricity may not fall on the same people.

[1 mark]

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10

Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

To what extent can renewable energy resources replace fossil fuels in meeting the energy needs of a modern society?

[25 marks]

OR

1 1 . 2

Soil is often described as a non-renewable resource. Discuss the causes and consequences of soil degradation and evaluate the strategies that can be used to manage soils sustainably.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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Do not write outside the box

END OF QUESTIONS